

# 30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

## Circles

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

### Question 1

**Tangents PA and PB from P have lengths  $(3x + 3)$  cm and  $(2x + 5)$  cm. Find x and each tangent length.**

Answer:

PA = PB, so  $3x+3=2x+5$ .  $x=2$ ; each tangent is 9 cm.

### Question 2

**Tangents PA and PB from P have lengths  $(3x + 4)$  cm and  $(2x + 7)$  cm. Find x and each tangent length.**

Answer:

PA = PB, so  $3x+4=2x+7$ .  $x=3$ ; each tangent is 13 cm.

### Question 3

**Tangents PA and PB from P have lengths  $(3x + 5)$  cm and  $(2x + 8)$  cm. Find x and each tangent length.**

Answer:

PA = PB, so  $3x+5=2x+8$ .  $x=3$ ; each tangent is 14 cm.

### Question 4

**Tangents PA and PB from P have lengths  $(3x + 1)$  cm and  $(2x + 7)$  cm. Find x and each tangent length.**

Answer:

PA = PB, so  $3x+1=2x+7$ .  $x=6$ ; each tangent is 19 cm.

### Question 5

**Tangents PA and PB from P have lengths  $(3x + 2)$  cm and  $(2x + 6)$  cm. Find x and each tangent length.**

Answer:

PA = PB, so  $3x+2=2x+6$ .  $x=4$ ; each tangent is 14 cm.

### Question 6

**Tangents PA and PB from P have lengths  $(3x + 6)$  cm and  $(2x + 9)$  cm. Find x and each tangent length.**

Answer:

PA = PB, so  $3x+6=2x+9$ .  $x=3$ ; each tangent is 15 cm.

### Question 7

**Tangents PA and PB from P have lengths  $(3x + 2)$  cm and  $(2x + 8)$  cm. Find x and each tangent length.**

Answer:

PA = PB, so  $3x+2=2x+8$ .  $x=6$ ; each tangent is 20 cm.

## Circles - Continued

### Question 8

Tangents PA and PB from P have lengths  $(3x + 5)$  cm and  $(2x + 9)$  cm. Find x and each tangent length.

Answer:

PA = PB, so  $3x+5=2x+9$ .  $x=4$ ; each tangent is 17 cm.

### Question 9

A point P is 13 cm from the centre of a circle of radius 5 cm. Find tangent length PT.

Answer:

OT is perpendicular to PT.  $PT = \sqrt{13^2 - 5^2} = 12$  cm.

### Question 10

A point P is 17 cm from the centre of a circle of radius 8 cm. Find tangent length PT.

Answer:

OT is perpendicular to PT.  $PT = \sqrt{17^2 - 8^2} = 15$  cm.

### Question 11

A point P is 25 cm from the centre of a circle of radius 7 cm. Find tangent length PT.

Answer:

OT is perpendicular to PT.  $PT = \sqrt{25^2 - 7^2} = 24$  cm.

### Question 12

A point P is 10 cm from the centre of a circle of radius 6 cm. Find tangent length PT.

Answer:

OT is perpendicular to PT.  $PT = \sqrt{10^2 - 6^2} = 8$  cm.

### Question 13

A point P is 15 cm from the centre of a circle of radius 9 cm. Find tangent length PT.

Answer:

OT is perpendicular to PT.  $PT = \sqrt{15^2 - 9^2} = 12$  cm.

### Question 14

A point P is 20 cm from the centre of a circle of radius 12 cm. Find tangent length PT.

Answer:

OT is perpendicular to PT.  $PT = \sqrt{20^2 - 12^2} = 16$  cm.

### Question 15

A point P is 29 cm from the centre of a circle of radius 21 cm. Find tangent length PT.

Answer:

OT is perpendicular to PT.  $PT = \sqrt{29^2 - 21^2} = 20$  cm.

### Question 16

A point P is 37 cm from the centre of a circle of radius 12 cm. Find tangent length PT.

Answer:

OT is perpendicular to PT.  $PT = \sqrt{37^2 - 12^2} = 35$  cm.

## Circles - Continued

### Question 17

**A radius to the point of contact is 3 cm. What is the angle between the radius and tangent?**

Answer:

The radius is perpendicular to the tangent, so the angle is 90 degrees.

### Question 18

**A radius to the point of contact is 4 cm. What is the angle between the radius and tangent?**

Answer:

The radius is perpendicular to the tangent, so the angle is 90 degrees.

### Question 19

**A radius to the point of contact is 5 cm. What is the angle between the radius and tangent?**

Answer:

The radius is perpendicular to the tangent, so the angle is 90 degrees.

### Question 20

**A radius to the point of contact is 6 cm. What is the angle between the radius and tangent?**

Answer:

The radius is perpendicular to the tangent, so the angle is 90 degrees.

### Question 21

**State the main formula, rule or theorem used in Circles.**

Answer:

The tangent at any point of a circle is perpendicular to the radius through that point.

### Question 22

**What do the important symbols or quantities in Circles represent?**

Answer:

Tangents drawn from the same external point are equal in length.

### Question 23

**Write the correct method for solving a Circles numerical.**

Answer:

Draw the radius to the point of contact, use the right angle, Pythagoras or equal-tangent property, then calculate.

### Question 24

**Which unit, sign convention or condition must be checked in a Circles answer?**

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

## Circles - Continued

### Question 25

**Why must the final answer in Circles be checked against the given condition?**

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

### Question 26

**What is one common error in Circles questions?**

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

### Question 27

**How should full working be shown in a Circles numerical?**

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

### Question 28

**When is a diagram, table or graph useful in Circles?**

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

### Question 29

**How can a student decide which formula to use in Circles?**

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

### Question 30

**What makes a high-scoring answer in Circles?**

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: CBSE Mathematics curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.